

Understanding and Reducing Customer Churn: A Data-Driven Retention Strategy

Final Stage 1 — Data Analytics Competition

Team: DataWizards

[EDIT: Member 1 — Member 2]

[EDIT: University]

Executive Summary

Customer churn is a direct threat to the premium income and profitability of a Texas auto insurer. Using data on 160,000 customers (overall churn rate 11.5%), we combined statistical driver analysis, survival analysis, interpretable regression and gradient-boosted machine learning to explain why and when customers leave and whom to prioritise. Churn risk is highest in tenure interval [90, 180) days. Our final model reaches an out-of-fold ROC-AUC of 0.706 (95% CI 0.701–0.710); targeting the 20% highest-risk customers captures 52% of all churners. Expected annual premium at risk is 17.50M USD (11.6% of the premium base). We recommend: (1) First-year onboarding programme; (2) Life-stage tailored engagement; (3) Investigate and act on 'cust orig date days since'. A campaign simulation suggests that targeting the top 10% of customers by expected loss yields a net benefit of 600.86K USD (ROI 1.5x) under our stated assumptions.

[EDIT: tambahkan 1 kalimat konteks spesifik dari soal (produk, pasar, tujuan manajemen) dan rapikan ringkasan ini maks. ½ halaman.]

1. Introduction and Business Problem

[EDIT: 1 paragraf konteks pasar & perusahaan dari soal. Untuk konteks industri Indonesia (penetrasi OJK, klaim surrender AAJI, regulasi PAYDI) pakai paragraf siap pakai di BAHAN_LATAR_BELAKANG.md §C.1.]

For a Texas auto insurer, every customer who leaves takes future premium income with them. The economics of retention are well documented: Reichheld and Sasser (1990) reported that reducing customer defections by 5% raised profits by 25%–85% across several service businesses, and in insurance the effect is sharper because acquisition costs are concentrated at the start of the contract, so policies that exit early often never recover them. Lapse also affects pricing and risk management, because products are priced under persistency assumptions (Eling & Kochanski, 2013). Not every exit is equally avoidable: Gottlieb and Smetters (2021) found that forgetting to pay explained 37.8% of recent life-insurance lapses, which suggests that part of churn can be addressed through process design rather than price.

This study answers four questions: (RQ1) Which customer, policy and service factors are associated with churn, and how strongly after controlling for the others? (RQ2) When during the customer lifecycle is churn risk highest? (RQ3) How accurately can churn be predicted, and which customers should be prioritised? (RQ4) Which interventions are most likely to reduce churn, and what is their expected economic value?

2. Data and Methodology

2.1 Data

The dataset contains 160,000 customers (plus 40,000 customers to be scored) and 20 original variables covering **[EDIT: demographics, policy characteristics, premium & payment behaviour, claims and service interactions]**. The target is Churn (churn = 1); the overall churn rate is 11.5%.

2.2 Data preparation

Data preparation included automatic type correction (e.g. numbers stored as text and date parsing), harmonisation of inconsistent category labels, removal of 0 duplicate records and treatment of impossible values. We engineered 12 domain features, including tenure (years), tenure band, new customer (<12 months), age band, premium-to-income ratio, cust orig date days since. Variables recorded only after churn (acct suspd date year, acct suspd date month, acct suspd date days since, acct suspd date) were excluded from modelling to prevent target leakage.

2.3 Analytical framework

Our framework has five layers (Figure 1): (1) univariate driver analysis with χ^2 and Mann-Whitney tests, effect sizes (Cramér's V, rank-biserial r), Information Value and Benjamini-Hochberg correction; (2) survival analysis (Kaplan-Meier, log-rank tests and Cox proportional hazards) to capture when churn occurs; (3) a multivariate logistic regression for interpretable odds ratios; (4) predictive machine learning (catboost, hgb, lgbm, logreg, xgb) evaluated with stratified 5-fold cross-validation, tuned with Bayesian optimisation (Optuna) and combined via ensemble selection, with model behaviour explained by SHAP values and partial dependence; and (5) doubly robust causal estimation (AIPW) to quantify the effect of controllable levers. Results were translated into business value with lift analysis, customer lifetime value, a risk-value matrix, a retention-campaign ROI simulation and what-if scenarios.

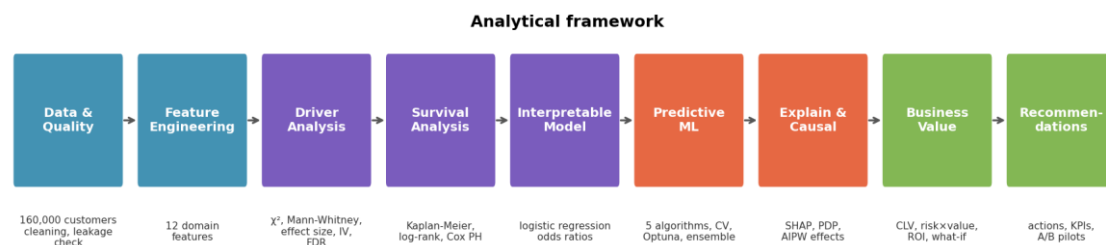


Figure 1. Analytical framework: from data preparation to evidence-based retention recommendations.

2.4 Validation

All performance figures are out-of-fold estimates. Target-dependent encodings were fitted inside each training fold. Sanity checks confirmed the absence of leakage (ROC-AUC with shuffled labels ≈ 0.5) and similar train/test distributions (adversarial validation AUC = 0.50). Uncertainty is reported with bootstrap 95% confidence intervals.

3. Findings

3.1 Churn overview

11.5% of customers churned, i.e. a **[EDIT: moderate/severe]** class imbalance; we therefore evaluate models with ROC-AUC and precision–recall rather than accuracy.

3.2 Key churn drivers

tenure band: customers in '0-6m' churn at 49.2% (4.3x the overall rate of 11.5%) versus 6.9% for '2-5y' (χ^2 p <0.001, Cramér's V = 0.39, IV = 0.93).

cust orig date month: churned customers have a median of 8.00 vs 9.00 for retained customers (weak/non-monotonic; rank-biserial r = -0.04, negligible effect, IV = 0.10). Highest-risk bin 6–7 churns at 20.5% (1.8x the average).

age band: customers in '25-34' churn at 13.9% (1.2x the overall rate of 11.5%) versus 8.0% for '65+' (χ^2 p <0.001, Cramér's V = 0.06, IV = 0.04).

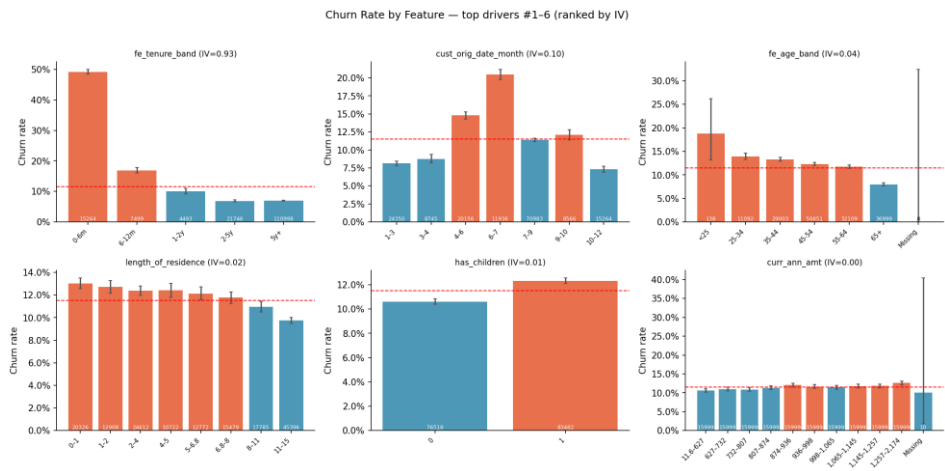


Figure 2. Churn rate by the strongest drivers (bars: churn rate with 95% CI; dashed line: overall churn rate).

Table 1. Driver evidence matrix (univariate, multivariate, survival and SHAP evidence).

Driver	High-risk group	Churn % (group)	Lift	IV	OR	HR	SHAP rank

tenure band	0-6m	49.20	4.27	0.93	12.63		10
age band	25-34	13.90	1.21	0.04	0.88	0.44	22
days tenure	22-191	47.70	4.14	0.89			1
tenure (years)	0.0602-0.523	47.70	4.14	0.89			2
cust orig date days since	22-191	47.70	4.14	0.89			3
length of residence	0-1	13.00	1.13	0.02			4

[EDIT: jelaskan 2-3 driver utama dengan bahasa bisnis — kenapa masuk akal di industri asuransi (lihat PLAYBOOK §D2).]

Multivariate evidence

Holding other factors constant, tenure band = 0-6m (vs '5y+') changes the odds of churn by +1163% (OR = 12.63, 95% CI 12.12-13.15, $p < 0.001$; average marginal effect +21.9 pp).

Holding other factors constant, age band = 65+ (vs '45-54') changes the odds of churn by -12% (OR = 0.88, 95% CI 0.84-0.93, $p < 0.001$; average marginal effect -1.1 pp).

3.3 When do customers churn?

Estimated probability of retaining a customer beyond 90 days: 98.6% (Kaplan-Meier).

Estimated probability of retaining a customer beyond 180 days: 95.2% (Kaplan-Meier).

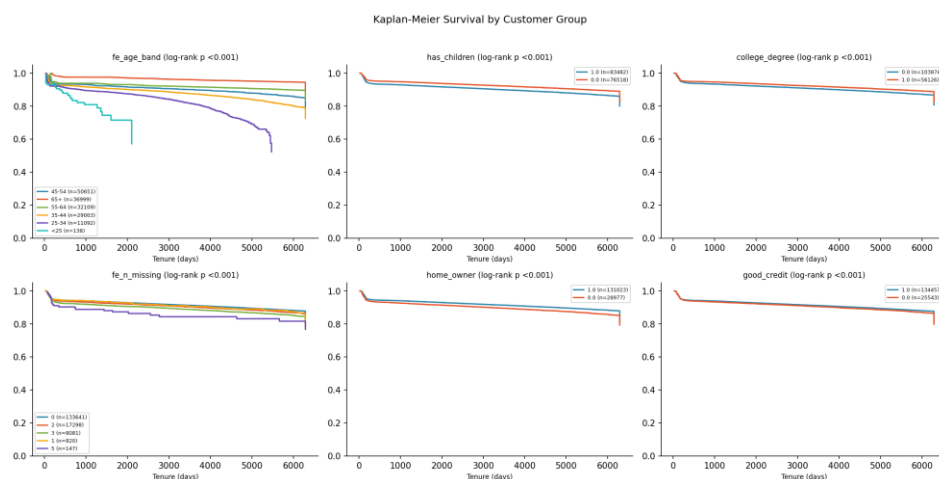


Figure 3. Kaplan-Meier survival curves by key customer groups (log-rank p -values in titles).

[EDIT: interpretasi — kapan momen kritis (onboarding? renewal pertama?) dan implikasinya untuk timing intervensi.]

3.4 High-risk segments

High-risk segment: IF tenure band = 0-6m AND length of residence $\leq$ 10.50 AND has children = 1 AND premium-to-income ratio $\leq$ 0.01 THEN churn rate = 52.8% (4.6x average; 3,258 customers, 2.0% of base, capturing 9.3% of all churners).

Table 2. Highest-risk and most loyal segments identified by a shallow decision tree.

Segment (decision-tree rule)	Customers	Churn %	Lift
tenure band = 0-6m AND length of residence $\leq$ 10.50 AND has children = 1 AND premium-to-income ratio $\leq$ 0.01	3258	52.80	4.58
tenure band = 0-6m AND length of residence $\leq$ 10.50 AND has children = 1 AND premium-to-income ratio $>$ 0.01	3403	51.00	4.44
tenure band $\neq$ 0-6m AND tenure band $\neq$ 6-12m AND tenure band $\neq$ 1-2y AND latitude $\leq$ 33.13	122559	6.90	0.60

3.5 Predictive model

The final model (mean_top3) achieves a cross-validated roc_auc of 0.7056 (ROC-AUC 0.7056) versus 0.50 for a random baseline.

At the selected threshold (0.23) the model identifies 44.9% of churners (recall) with a precision of 49.3%, i.e. 4.3x better targeting than random selection.

Final model ROC-AUC = 0.706 (95% bootstrap CI 0.701–0.710).

Table 3. Cross-validated model comparison.

Model	OOF roc_auc	CV std	ROC-AUC	PR-AUC
lgbm	0.70	0.00	0.70	0.34
lgbm_multiseed	0.70	0.00	0.70	0.33
xgb_tuned	0.70	0.00	0.70	0.33
lgbm_tuned	0.70	0.00	0.70	0.33

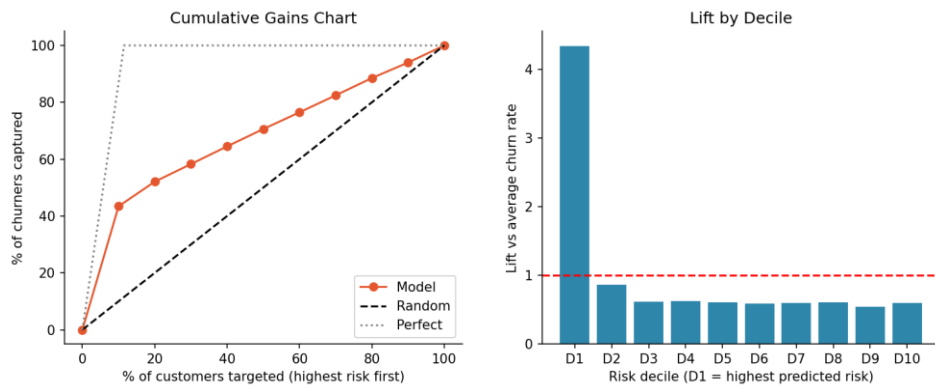


Figure 4. Cumulative gains and lift by risk decile: the model concentrates churners in the highest-risk deciles.

3.6 Explaining churn risk

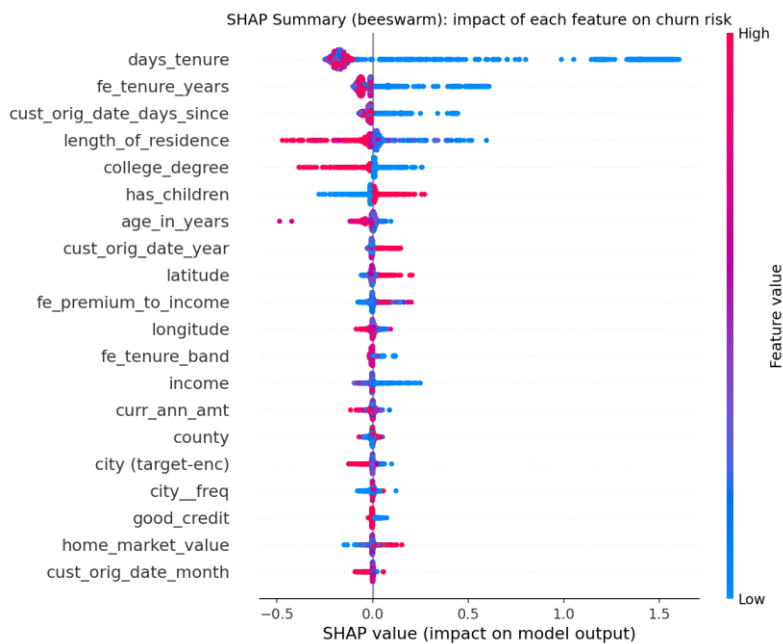


Figure 5. SHAP summary: each dot is a customer; positive values increase churn risk (colour = feature value).

Partial dependence: moving days tenure from 119.00 to 6.29K changes average predicted churn from 30.9% to 7.9%.

Partial dependence: moving tenure (years) from 0.33 to 17.22 changes average predicted churn from 14.4% to 10.0%.

[EDIT: hubungkan SHAP dengan temuan statistik — driver mana yang konsisten di semua metode (robust).]

4. Business Impact and Recommendations

4.1 Value at risk and prioritisation

Targeting the top 20% highest-risk customers captures 52.1% of all churners (2.6x lift over random targeting); the top decile churns at 50.0% vs 6.9% in the bottom decile. KS = 0.40.

The High-risk tier (20% of customers, $p \geq 0.08$) has an actual churn rate of 30.0% and contains 52.1% of all churners.

Expected annual premium at risk from churn is 17.50M USD (11.6% of the annual premium base); the High-risk tier alone accounts for 49.8% of it.

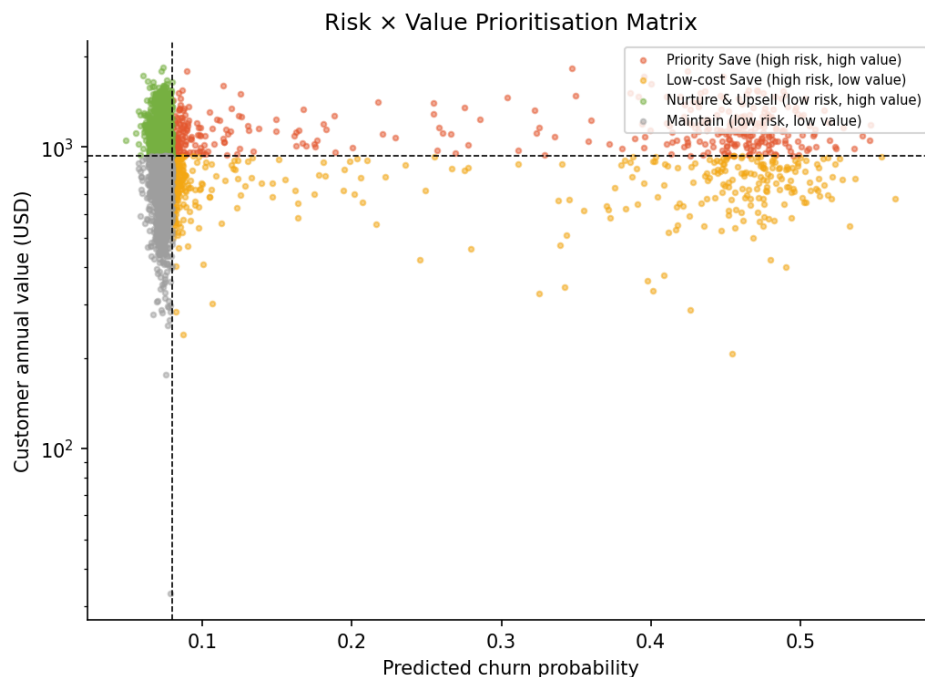


Figure 6. Risk $\times$ value matrix: 'Priority Save' customers combine high churn risk with above-median value.

4.2 Retention campaign simulation

Simulated retention campaign (assumed success rate 25%, cost 25.00 USD per contact): targeting the top 10% ranked by 'Model: expected loss ($p \times \text{value}$)' maximises net profit at 600.86K (ROI 1.5x), versus -178.30K for random targeting of the same size.

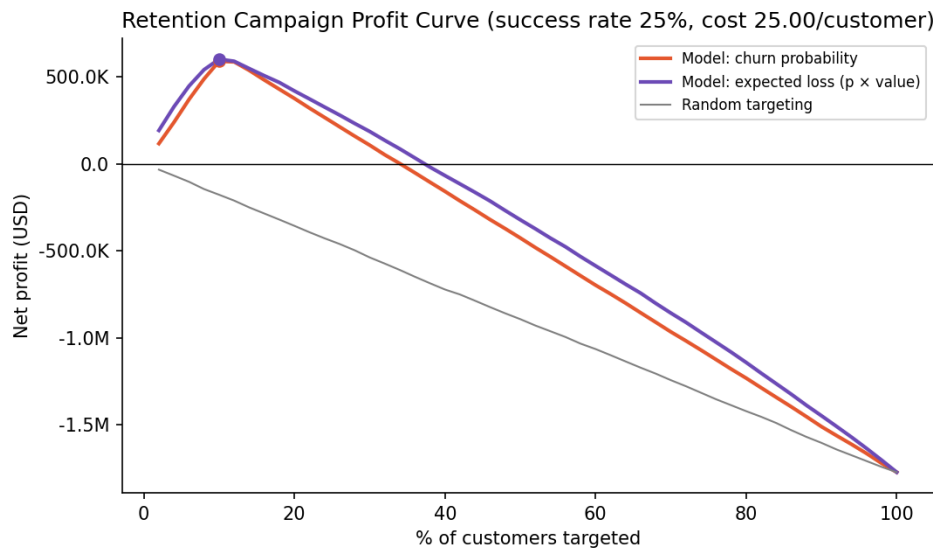


Figure 7. Net profit of a retention campaign by share of customers targeted (model vs random targeting).

[EDIT: sebutkan asumsi bisnis — margin 15%, cost per contact 25.00 USD, success rate 25% — dan sumbernya (soal / asumsi wajar); rujuk tabel sensitivity di Appendix.]

4.5 Churn personas

Persona P2 (82% of at-risk customers, actual churn 7.2%): risk driven mainly by no dominant driver (moderate, diffuse risk).

Persona P1 (18% of at-risk customers, actual churn 46.0%): risk driven mainly by days tenure (+1.34); tenure (years) (+0.39); cust orig date days since (+0.19).

[EDIT: beri nama tiap persona (misal 'Price-shocked renewers', 'Service-frustrated claimants') + 1 intervensi khusus per persona.]

4.6 Recommendations

Table 4. Evidence-based retention recommendations.

#	Recommendation	Evidence	Expected impact	KPI
1	First-year onboarding programme	'0-6m' churns at 49.2% (4.3x avg); IV 0.93; OR 12.63; SHAP rank #10		13th-month persistency ratio; first-year churn rate
2	Life-stage tailored engagement	'25-34' churns at 13.9% (1.2x avg); IV 0.04; OR 0.88; HR 0.44; SHAP rank #22		Churn rate by age band; adoption of tailored offers

3	Investigate and act on 'cust orig date days since'	'22–191' churns at 47.7% (4.1x avg); IV 0.89; SHAP rank #3		Churn rate of the 'cust orig date days since' high-risk group
4	Household-based cross-sell and life-event triggers	'0–1' churns at 13.0% (1.1x avg); IV 0.02; SHAP rank #4		Products per household; churn after life events
5	Deploy a churn early-warning system	Model ROC-AUC 0.706; top-20% captures 52% of churners	optimal campaign: target top 10% → net 600.86K	Precision/recall of alerts; retained revenue vs control group
6	Validate with controlled pilots (A/B tests)	What-if results are associational		Uplift in retention rate (treated – control)

1. First-year onboarding programme. Welcome call, policy walkthrough, app activation and 90-day check-in for new customers; first renewal treated as a critical moment.
2. Life-stage tailored engagement. Tailor products, pricing communication and service channels to the highest-risk age groups (e.g. digital-first onboarding and flexible payment for younger customers; loyalty recognition for long-standing older customers).
3. Investigate and act on 'cust orig date days since'. Customers with high-risk values of 'cust orig date days since' churn more; design a targeted intervention and test it in a pilot.

[EDIT: tambahkan roadmap — Quick wins (0–3 bulan), Medium (3–6 bulan), Long term (6–12 bulan) — dan rencana pengukuran (A/B test dengan control group).]

5. Conclusion and Limitations

Churn at a Texas auto insurer is driven primarily by **[EDIT]**, and is concentrated in identifiable segments and lifecycle stages. A validated predictive model (ROC-AUC of 0.706 (95% CI 0.701–0.710)) enables proactive, value-based retention targeting. **[EDIT: 1 kalimat jawaban langsung untuk tiap RQ1–RQ4, dan kaitkan 1–2 temuan dengan literatur — pilih kalimat dari BAHAN_LATAR_BELAKANG.md §C.5 yang sesuai temuanmu.]**

Limitations: (i) the data are a single snapshot, so relationships are associational rather than causal; (ii) the ROI simulation relies on assumed costs and success rates (see sensitivity analysis); (iii) potentially important factors such as competitor offers and life

events are not observed. Future work should validate interventions through randomised pilots and enrich the data with **[EDIT: interaction logs / stated reasons for leaving]**.

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Appendix (not counted in the page limit)

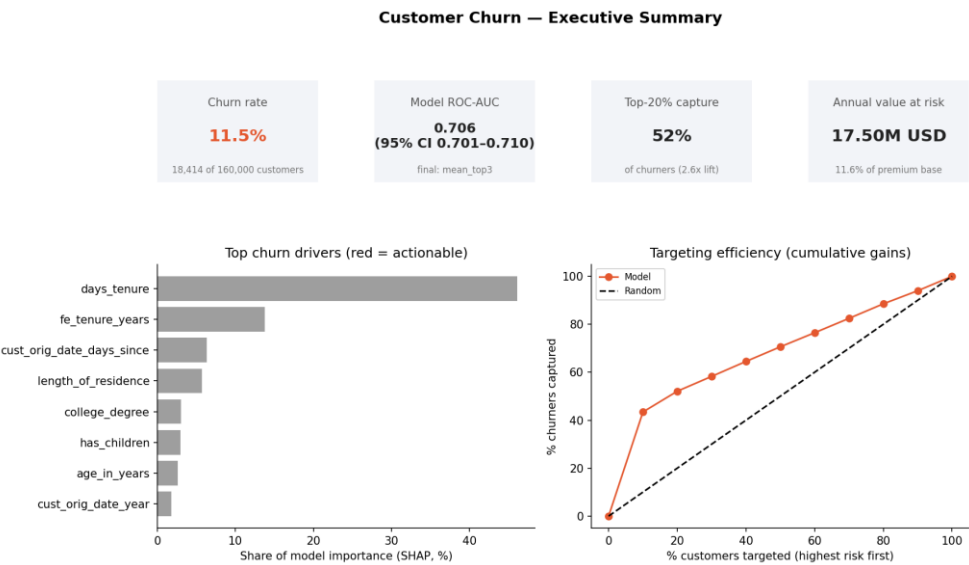


Figure 8. Executive summary dashboard.

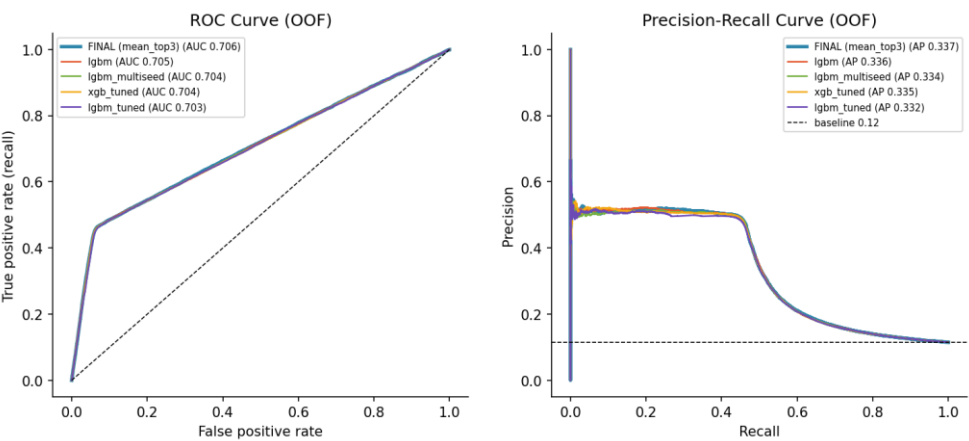


Figure 9. ROC and precision-recall curves (out-of-fold).

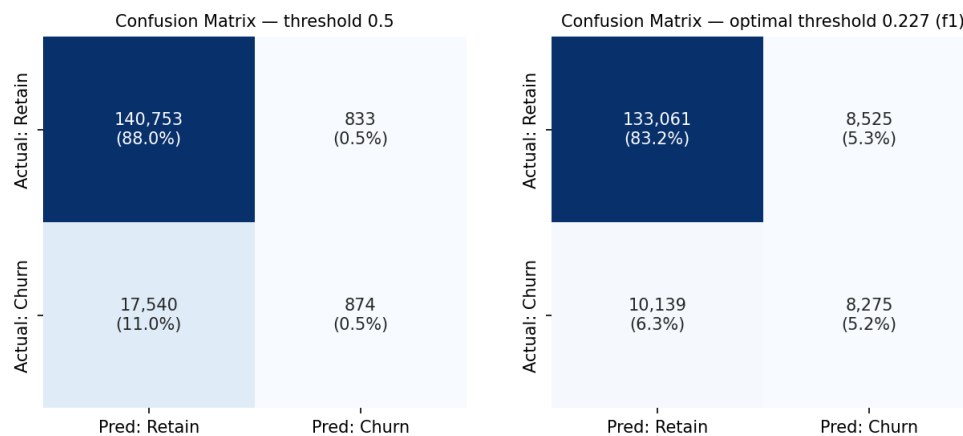


Figure 10. Confusion matrices at 0.5 and at the optimised threshold.

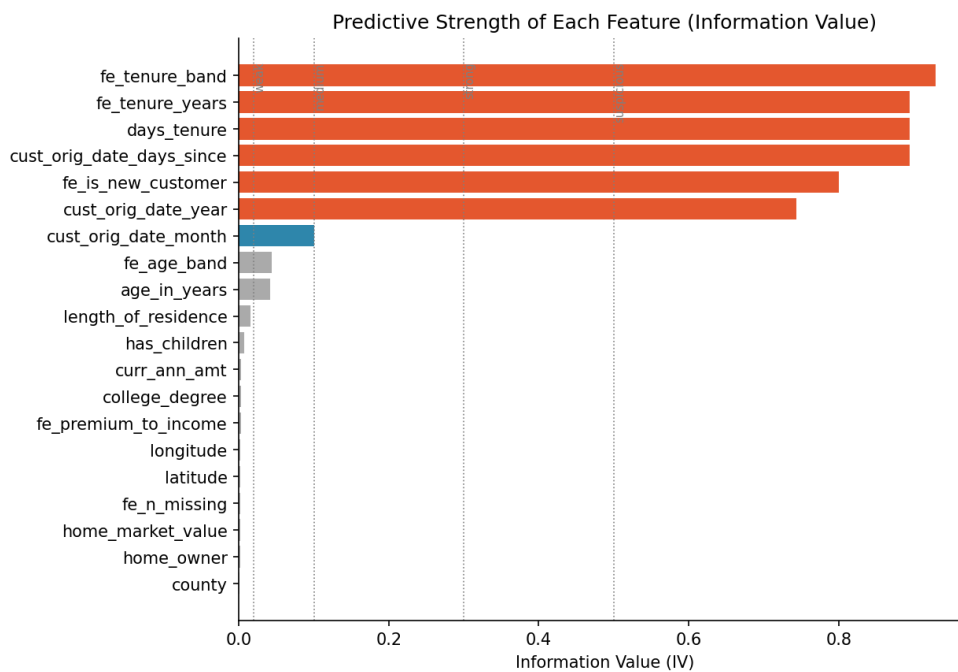


Figure 11. Information Value of each feature.

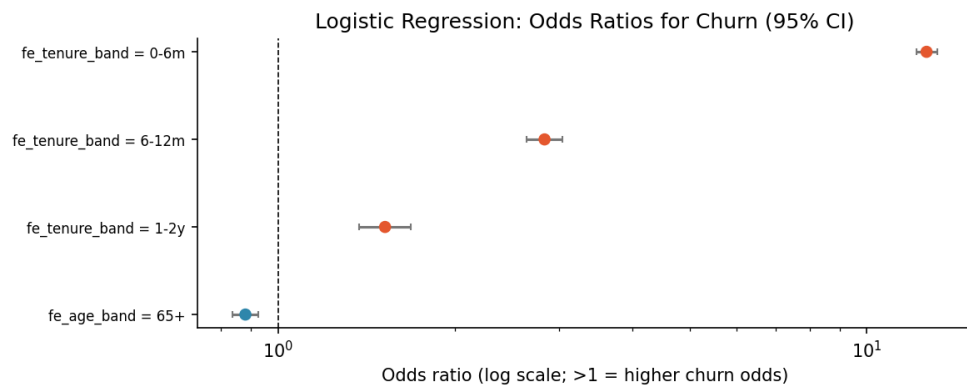


Figure 12. Logistic regression odds ratios (95% CI).

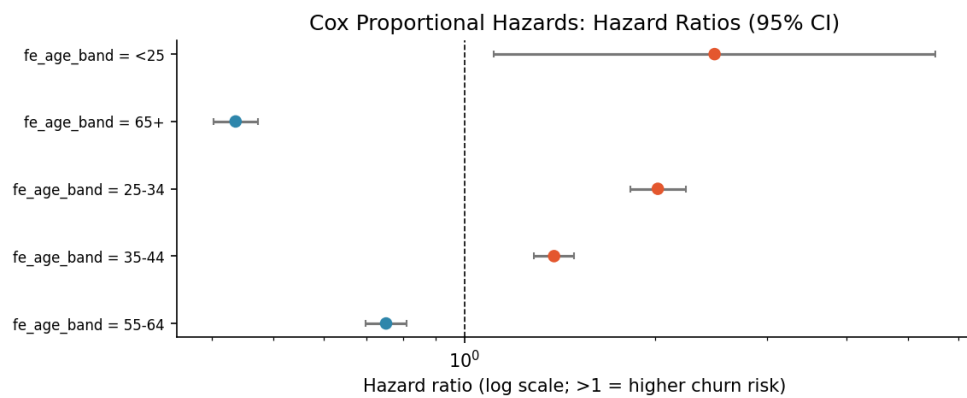


Figure 13. Cox proportional hazards: hazard ratios (95% CI).

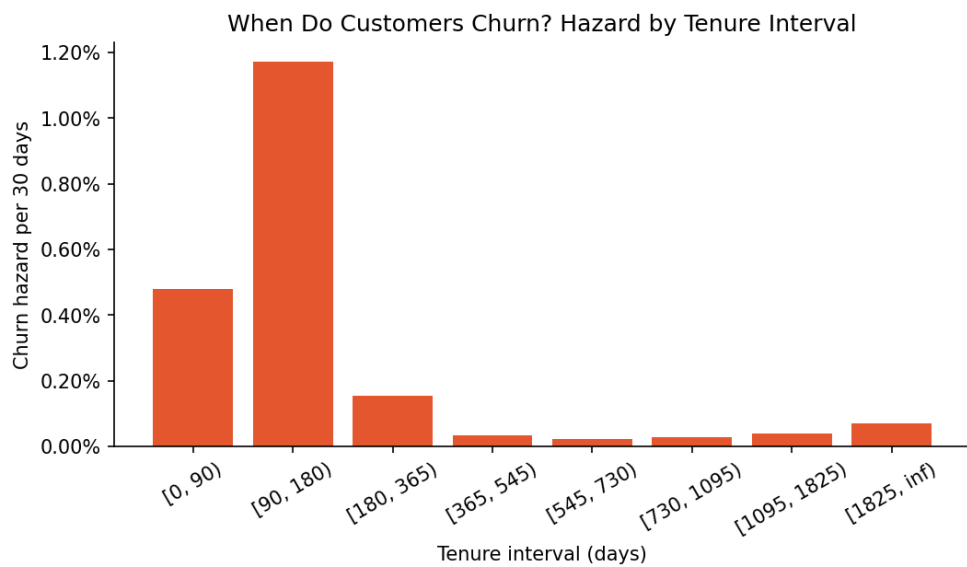


Figure 14. Churn hazard by tenure interval.

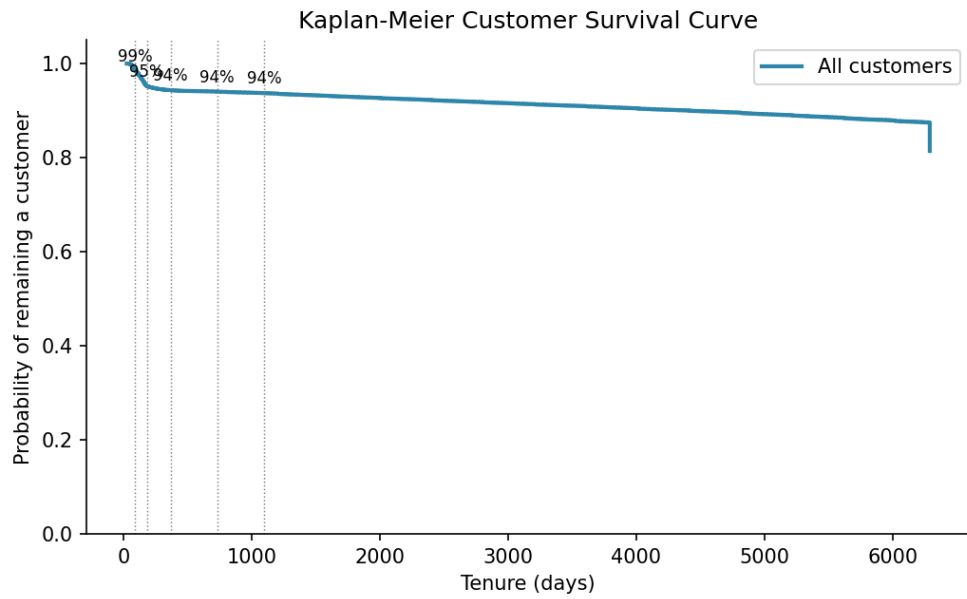


Figure 15. Overall Kaplan-Meier curve.

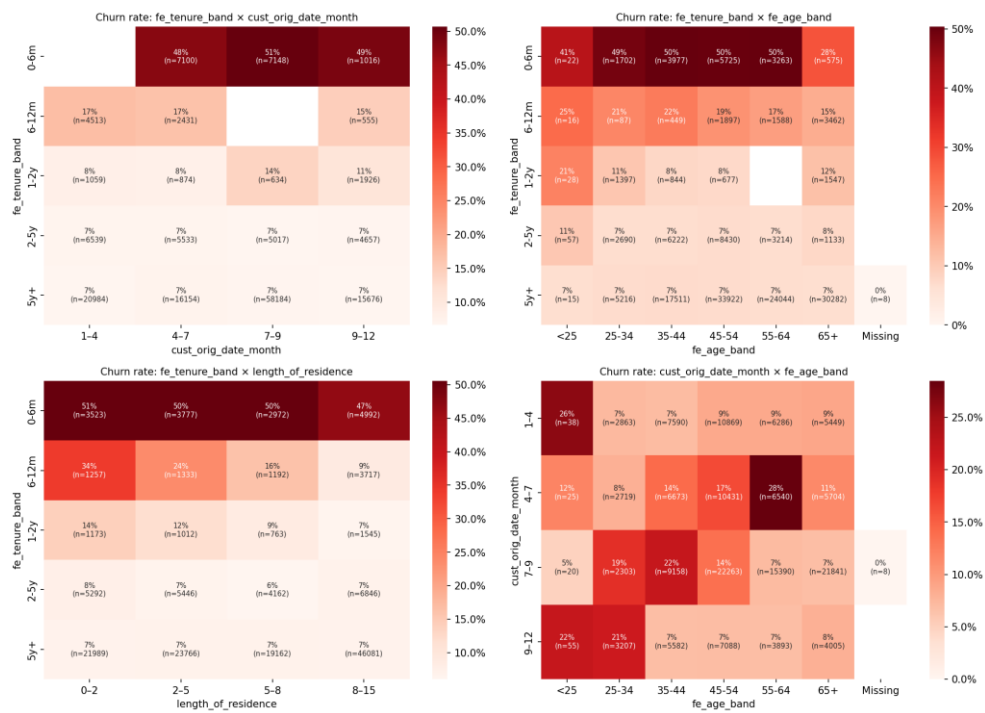


Figure 16. Two-way churn-rate heatmaps.

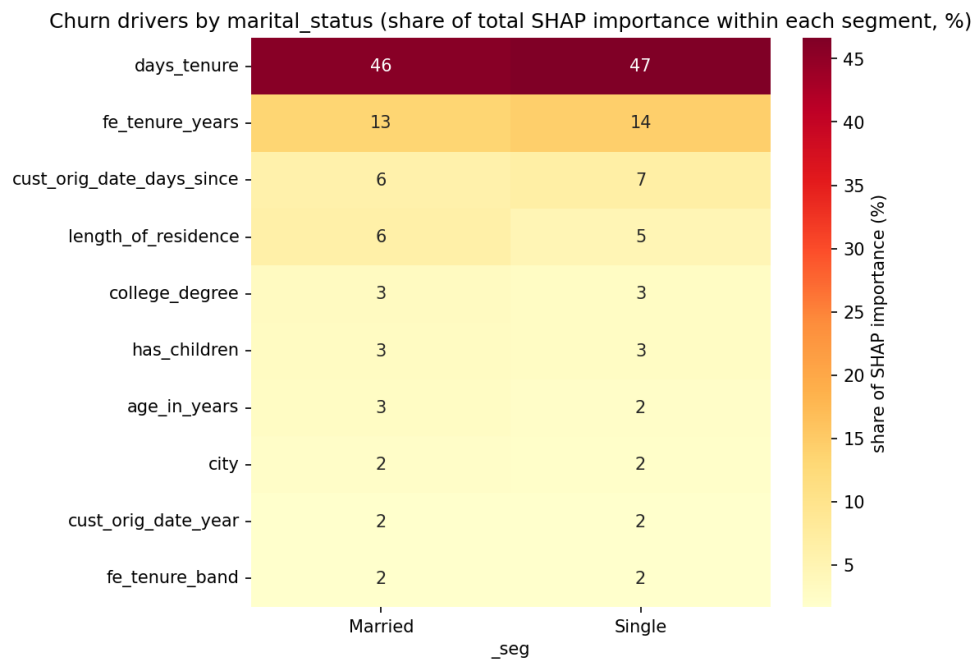


Figure 20. Driver importance by segment.

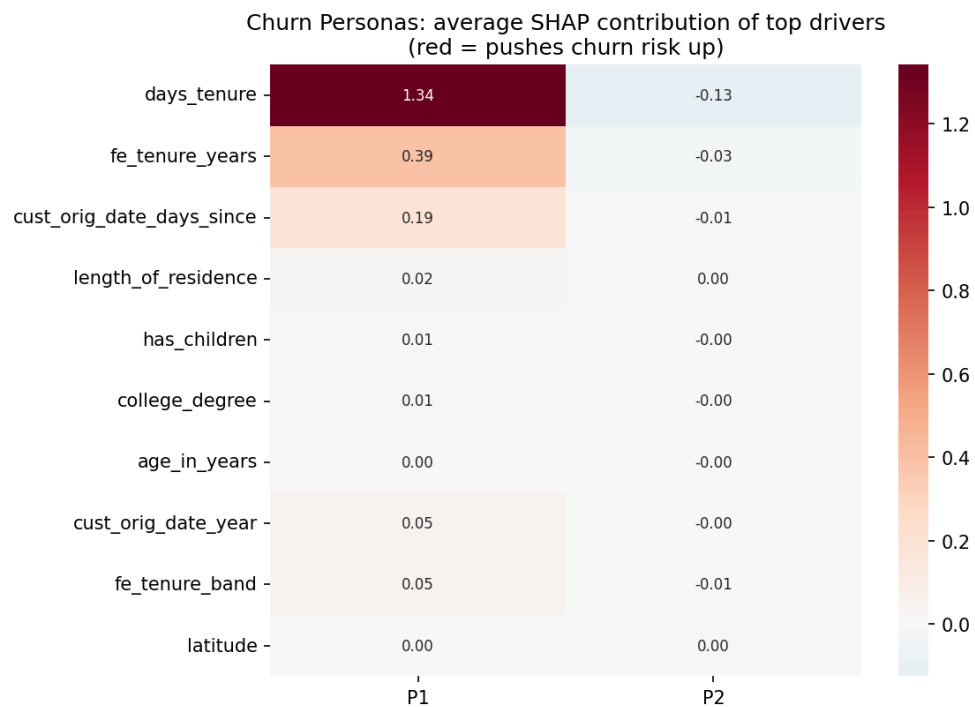


Figure 21. Churn personas (SHAP clustering).

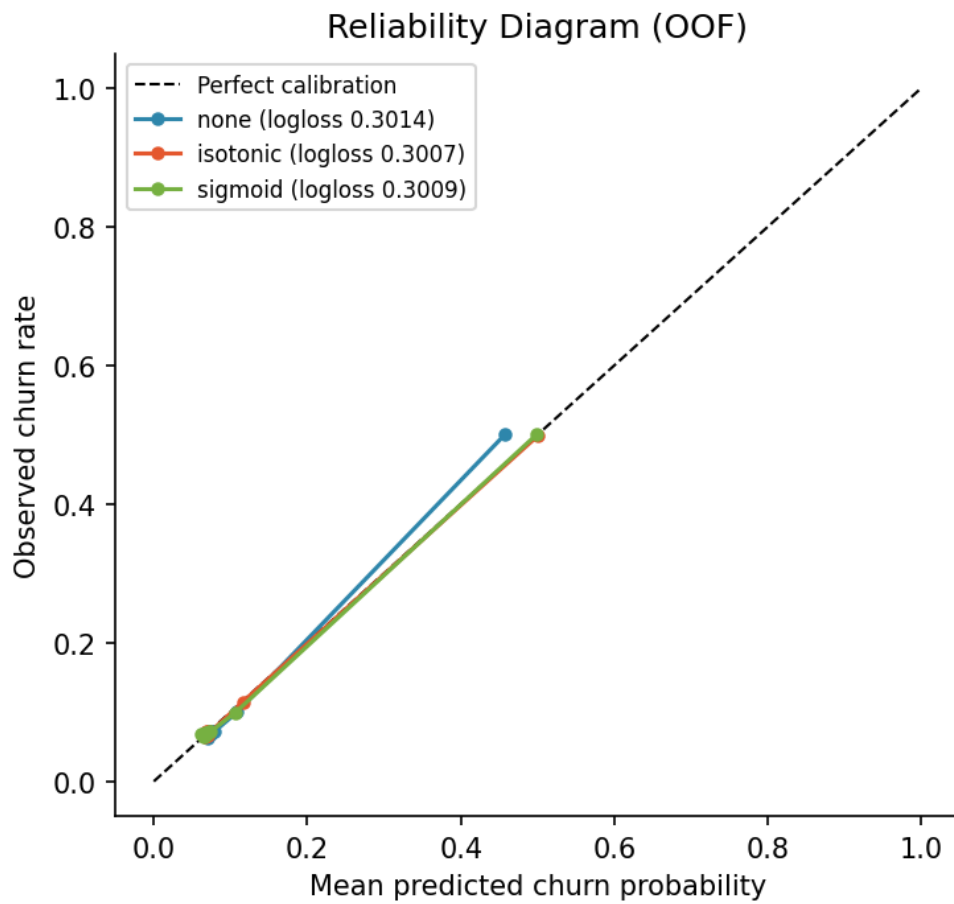


Figure 22. Calibration (reliability) diagram.

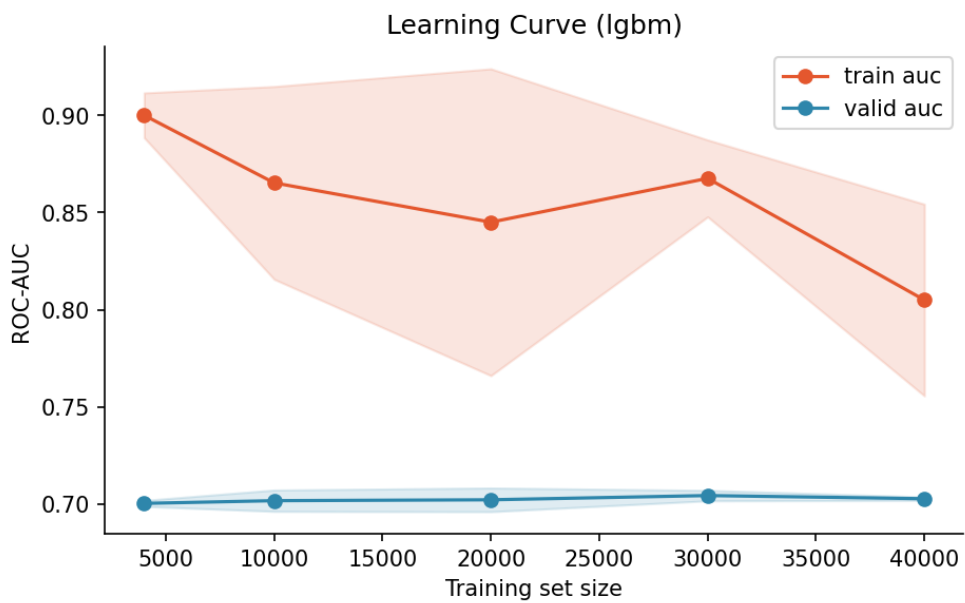


Figure 23. Learning curve.

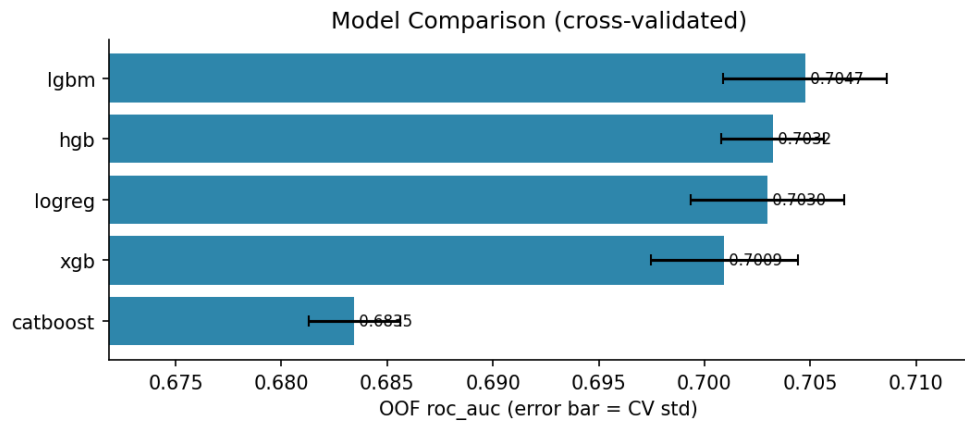


Figure 24. Model comparison chart.

Table 5. Risk tiers.

tier	n_train	actual_churn_rate	avg_predicted	share_churners_%	n_test
High	32000	0.30	0.28	52.09	7054
Medium	48000	0.07	0.08	18.52	10468
Low	80000	0.07	0.07	29.40	22478

Table 6. Campaign sensitivity analysis.

success_rate	cost_per_contact	optimal_target_%	max_net_profit	roi
0.12	12.50	10.17	301,878.61	1.49
0.12	25.00	6.08	103,021.31	0.42
0.12	50.00	2.00	-24,358.09	-0.15
0.25	12.50	14.25	827,143.43	2.90
0.25	25.00	10.17	603,757.23	1.49
0.25	50.00	6.08	206,042.63	0.42
0.38	12.50	55.08	1.52M	1.38
0.38	25.00	10.17	1.11M	2.73
0.38	50.00	10.17	702,310.84	0.86

Table 7. Churn persona profiles.

persona	n_in_sample	share_of_at_risk_%	avg_pred_churn	actual_churn_rate	top_drivers	tenure band	age band
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P2	851	81.83	0.08	0.07	no dominant driver (moderate, diffuse risk)	5y+	45-54
P1	189	18.17	0.45	0.46	days tenure (+1.34); tenure (years) (+0.39); cust orig date days since (+0.19)	0-6m	45-54

Table 8. Paired bootstrap comparison of models.

final_vs	diff_roc_auc	ci_low	ci_high	p_value_one_sided	significant
lgbm	0.00	-0.00	0.00	0.23	False
lgbm_multiseed	0.00	-0.00	0.00	0.12	False
xgb_tuned	0.00	-0.00	0.00	0.07	False
lgbm_tuned	0.00	-0.00	0.01	0.10	False
logreg	0.00	-0.00	0.01	0.09	False